UE24CS352A: MACHINE LEARNING MINI-PROJECT REPORT

Vision-Based Precision Pose Estimation for Autonomous Formation Flying

Team: Manish Bala & Manasvi D R | **PES University**

Repo: github.com/manishribil/Vision-Based-Pose-Estimation

1. PROBLEM STATEMENT & MOTIVATION

Autonomous multi-UAV formation flight is vital for cooperative search and rescue, perimeter monitoring, and collaborative cargo transport. Conventional swarms rely on Global Navigation Satellite Systems (GNSS) and inter-drone radio telemetry, which are acutely vulnerable to RF jamming, spoofing, multipath degradation, and complete denial in indoor or contested environments. This project develops a **passive onboard monocular vision pipeline** that detects peer drones, estimates 2D centroids and 3D relative pose, and continuously monitors formation health without transmitting radio signals.

2. DATASET SPECIFICATIONS

Trained and benchmarked on the Roboflow *Multiple Drones* aerial formation dataset:

- **Target Drone:** 3DR Solo quadcopters (physical diagonal width W = 0.46 m).
- **Resolution:** Native 640 × 640 RGB imagery captured from chase-drone perspectives.
- **Test Split Scope:** 96 sequential flight frames with 277 annotated ground-truth drone instances.
- **Dynamics:** Scale variation (box widths 8.4 px to 154.6 px), dynamic cloud/sky contrast, and multi-drone occlusion.

3. METHODOLOGY & 10-PHASE ARCHITECTURE

Phase 1 & 2 (Detection & Centroid Extraction): YOLO11n outputs bounding boxes [x1, y1, x2, y2]. Drone centroids are computed as *c_x = (x1+x2)/2*, *c_y = (y1+y2)/2*.

Phase 3 (Spatial Relations): Calculates horizontal and vertical relative bearing between all pairs: *dx = x_B - x_A*, *dy = y_B - y_A*.

Phase 4 (Evaluation): Benchmark validation on test split (Precision, Recall, mAP).

Phase 5 & 6 (Normalization & Mapping): Centroids are normalized as *x_norm = c_x / W_img*, *y_norm = c_y / H_img* to ensure scale and resolution invariance across cameras.

Phase 7 & 8 (Pairwise Euclidean Spacing): Evaluates pairwise normalized separation *dist = sqrt(dx_norm^2 + dy_norm^2)* across all pairs and summarizes swarm statistics.

Phase 9 (Distribution Visualization): Visualizes formation density charts.

Phase 10 (Monocular 3D Pose & Formation Health Tracking):
• *Pinhole Depth Estimation:* **Z = (f · W) / w**, where *w* is detected box width in pixels, *W = 0.46 m*, and *f = 554 px* (assumed 60° HFOV).
• *Lateral Displacements:* **X = (c_x - c0) · Z / f**, **Y = (c_y - c0) · Z / f**.
• *Dimensionless Spacing:* **s = d_px / w_bar** (in drone-widths). Classified into: **Maintained** (3.0 ≤ s ≤ 10.0), **Too Close** (s < 3.0), or **Too Far** (s > 10.0).
• *Live Telemetry HUD & Radar:* Real-time status badge and top-down X-Z radar view.

FORMATION TRACKING & TELEMETRY HUD

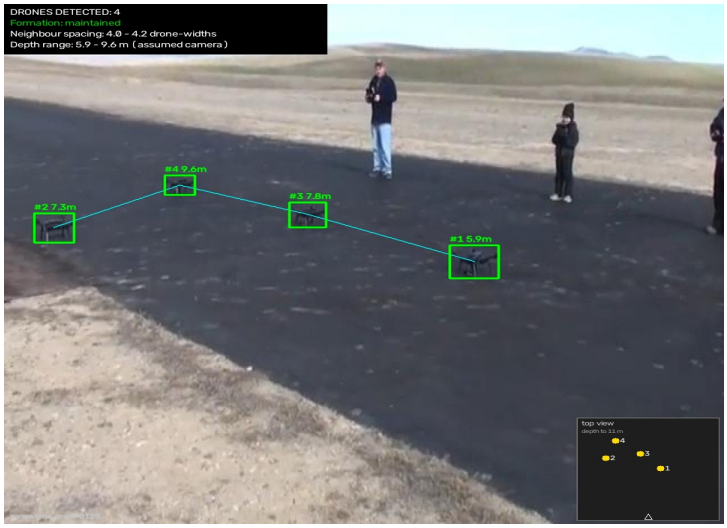


Figure 1: Phase 10 Output showing 4 tracked drones with 3D depths, inter-vehicle links, telemetry HUD banner, and top-down radar view inset.

SYSTEM PIPELINE STAGES

Step 1: Raw Video Stream: 640x640 Test Video Frames / Real-Time Chase Camera
Step 2: YOLO11n Detector: High-Recall Localization (2.6M parameters, >60 FPS)
Step 3: Centroid Extraction: Centroids (c_x, c_y) & Bearing Relationships
Step 4: Scale Normalization: Resolution-Invariant Formation Geometry Mapping
Step 5: 3D Monocular Range: Pinhole Depth Z = (f·W)/w | Lateral Offsets X, Y in meters
Step 6: Formation Health Engine: Dimensionless Nearest-Neighbor Spacing (Drone-Widths)
Step 7: Telemetry HUD & Radar: Live Status Badge & Bird's-Eye Top-Down Projection

4. DETECTION EVALUATION RESULTS

Metric	Score	Significance
Precision	**99.63%**	Near-zero false alarms in operating domain
Recall	**99.51%**	99.5% of drones detected across all test frames
mAP @ 0.50	**99.36%**	Outstanding localization at standard threshold
mAP @ 0.50:0.95	**85.12%**	High strictness across small & large scales
Test Dataset Scope	**96 frames**	277 ground-truth annotated drone targets

5. DEPTH ERROR & MATHEMATICAL SENSITIVITY

Ground-Truth Comparison (272 matched detections, mean IoU = 0.915):
• **MAE:** 0.252 m | **RMSE:** 0.360 m | **Mean Error (MAPE):** 2.5% | **90th Pct:** 5.2%.
Mathematical Sensitivity Derivation:
Differentiating $Z = (f \cdot W) / w$ with respect to pixel box width w:

$$dZ / dw = - (f \cdot W) / w^2 = - Z / w$$

In relative fractional terms: **| δZ / Z | = | δw / w |**
Physical Implication: Percentage depth error equals percentage box width error. For near drones (*w = 120 px*), a 1 px jitter causes only **0.83%** depth error. For distant drones (*w = 10 px*), 1 px jitter causes **10.0%** error. Thus, bounding-box regression precision dictates far-field range uncertainty.

SENSITIVITY & ERROR DISTRIBUTION

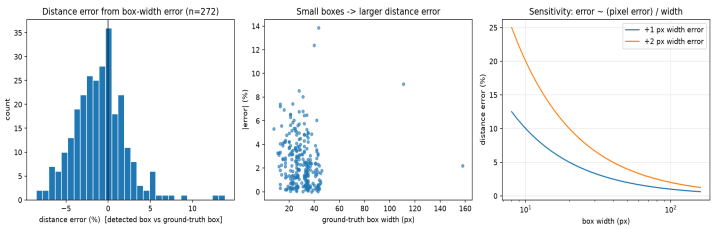


Figure 2: Ground-truth distance error distribution and sensitivity curve (|error| ~ 1/w).

6. EDGE-CASE STRESS TESTING & FAILURE MODES

Condition	Conf	Empirical Behavior / Impact
Motion Blur (K=15)	0.11	Significant confidence drop due to rotor smear
Motion Blur (K=35)	0.00	Complete detection dropout (0 boxes found)
Darkness (50% & 20%)	**0.35**	High silhouette contrast against sky preserves hits
Boundary Crop (35%)	0.31	Truncated width artificially inflates estimated depth
Urban Scene (bus.jpg)	0.47	False positive on window textures (dataset sky bias)
White Drone (drone.jpg)	0.39	Split partial boxes due to dark 3DR Solo training bias

7. FORMATION STATUS SUMMARY (96 TRAJECTORY FRAMES)

• **Maintained Formation:** 44 frames (45.8%)
• **Too Far (Outside Limit):** 34 frames (35.4%)
• **Single Drone Tracked:** 12 frames (12.5%)
• **Too Close (Collision Hazard):** 6 frames (6.3%)
• **Nearest-Neighbor Spacing:** Median 5.9 drone-widths (IQR 4.1 - 9.4)
• **Depth Extent:** 1.6 m to 30.2 m (Median 8.5 m)

8. CONCLUSIONS & EQUITABLE INDIVIDUAL CONTRIBUTIONS

Conclusions & Practical Takeaways: The project proves that high-precision relative pose estimation for autonomous formation flight is achievable via lightweight onboard monocular vision. Spacing in dimensionless drone-widths provides an immediate, calibration-free collision avoidance signal. Physical metre distance estimation is viable with known vehicle airframe width (MAE = 0.252 m). Future work includes EKF temporal filtering to eliminate far-field pixel jitter and negative-sample fine-tuning to eliminate urban false positives.

Individual Contributions:
• **Manish Bala:** YOLO11n object detector training & validation (Phases 1, 4); 2D centroid extraction pipeline (Phase 2); spatial bearing and orientation analysis (Phase 3); coordinate normalization & initial distance analysis (Phases 6, 7); core git repository setup.
• **Manasvi D R:** Pinhole monocular 3D distance and lateral displacement formulation (Phase 10); nearest-neighbor dimensionless spacing formulation; dynamic formation health classifier (Maintained / Too Close / Outside Limit) & top-down radar projection; ground-truth error sensitivity & box jitter mathematical derivation (|δZ/Z| = |δw/w|); edge-case robustness test suite (*tools/edge_case_tests.py*); real camera focal calibration & error analysis tool (*tools/manasvi_distance_viz.py*); end-to-end master verification harness (*verify_pipeline.py*) and executive PDF report generator.